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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 11/27/2005 |
| **PI Name:** | Muday, Gloria Ph.D. | | |
| **Project Title:** | Regulation of Auxin Transport by Phosphorylation and Flavonoids during Gravitropism in Arabidopsis | | |
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Although observations of the ability of plants to change their growth and developmental patterns in response to gravity date back to Darwin, the mechanisms which control this process are still not understood. One requirement for gravitropic bending is the redistribution of the plant hormone auxin. The potential for understanding how auxin transport is regulated during gravitropic bending has increased dramatically with the isolation of Arabidopsis mutants with defects in genes encoding both auxin transport proteins and regulatory proteins. The goal of this proposal will be to examine two potential mechanisms for regulation of auxin transport during gravitropism. The experimental approach will combine the use of Arabidopsis plants with mutations in genes that alter the regulation of auxin transport and gravity response and inhibitor treatments that mimic these mutant phenotypes; transgenic plants that provide information on changes in gene expression; and computer aided analysis of plant gravity response.
The proposed experiments will examine synthesis of compounds that act to change the activity of auxin transporters and

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| **Task Description:** | examine alterations in phosphorylation states that regulate auxin transport during plant gravity response. Preliminary evidence indicates that mutants that synthesize no flavonoids or that have reduced phosphatase activity exhibit significant delays in gravitropic bending. Both in vitro and in vivo evidence indicate that flavonoids are natural regulators of auxin transport. The localization of expression of flavonoid biosynthetic enzymes and flavonoid accumulation will be examined in concert with measurements of gravitropism to determine if the amount or distributions of these compounds affect gravitropic bending. Additionally, the role of phosphorylation in regulation of auxin transport will be examined using Arabidopsis plants with mutations that affect phosphatase activity. Finally, the possibility that phosphorylation and flavonoid synthesis are interconnected regulatory mechanisms will be examined. Phosphorylation states of cellular proteins may affect the transcription of genes encoding flavonoid biosynthetic enzymes or flavonoids may act as kinase inhibitors. Together these experiments should provide a clear sense of how flavonoids and phosphorylation act to regulate auxin transport and why alterations in either flavonoid synthesis or phosphorylation reduce the ability of plants to respond to gravity. The goal of this proposal will be to examine two potential mechanisms for regulation of auxin transport during gravitropism. The proposed experiments will examine synthesis of compounds that act to change the activity of auxin transporters and examine alterations in phosphorylation states that regulate auxin transport during plant gravity response. The proposed and completed experiments provide new insight into the mechanisms by which plants control their growth and response to gravity. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The goal of the research supported by this grant is to increase understanding of a basic biological process, the response of plants to gravity. This study focuses on the role of transport of a class of plant hormones, the auxins, in gravity response and explores how this transport affects plant growth and development. This study will also examine polarity in the distribution of proteins, polarity in cells and tissues, and the polar movement of auxin in order to determine the relationship between these processes and the changes in these processes during gravity response. The long-term goals of this study include understanding how one of the proteins which controls auxin transport functions and how gravity stimuli may affect this protein's function to allow changes in auxin transport and changes in growth in response to gravity. A clearer understanding of the mechanisms by which plants respond to changes in the vector of gravity may provide important insight into predicting how plants will grow in the absence of gravity and may facilitate the design of experiments to study plant response during space flight. |
| **Task Progress:** | Regulation of Auxin transport by flavonoids during gravitropism:<br>We explored the role of flavonoids as endogenous regulators of auxin transport and asking if these molecules facilitate the redistribution of auxin in response to gravitropic stimulation. The tt4 mutant, which has a mutation in a gene encoding a flavonoid biosynthetic enzyme, makes no flavonoids. Auxin transport is elevated in multiple tissues in this mutant, consistent with flavonoids acting as endogenous negative regulators. Charles demonstrated that there is elevated movement of auxin from the root tip back in the flavonoid deficient mutant, consistent with flavonoids regulating this polarity of auxin transport, which has been linked to gravitropic responses (Buer and Muday, 2004).<br><br>We characterized the gravity response in the tt4 mutant in detail. The roots of tt4 are much slower in initiating gravitropic bending. This defect in gravitropism can be reversed by feeding the tt4 plants naringenin, a flavonoid precursor that bypasses the step with the missing enzyme (Buer and Muday, 2004). Since the absence of flavonoids delays gravitropic bending, we asked whether gravitropic reorientation results in changes in accumulation or localization of flavonoids that may alter auxin transport at the root tip. This experiment is possible using a compound that becomes fluorescent in response to binding of flavonoids. We found that flavonoid localization does not change in response to gravity, but the concentration of flavonoids is increased in the root epidermal cells, in which gravitropic bending occurs (Buer and Muday, 2004). The model best supported by these combined results, is that flavonoids are locally induced at the root tip in response to gravity stimulation and that these compounds act to prevent polar auxin transport in the root, facilitating the formation of lateral gradients of auxin in response to the gravitropic signal.<br><br>Phosphorylation control of auxin transport during gravity response:<br><br>A second major avenue of investigation has been to explore the regulation of auxin transport by phosphorylation and to ask if this is a mechanism that allows changes in auxin transport in response to gravity. We have focused our studies on two mutants with altered phosphorylation signaling. The rcn1 mutant, which has a defect in a protein phosphatase regulatory subunit, and the pinoid (pid9) mutant, which has a defect in a kinase that has been suggested to regulate auxin transport and/or auxin signaling. Our results have indicated that both mutants have defects in polar auxin transport in roots and altered gravitropic bending. Our results with rcn1 suggest that elevated auxin transport can swamp out formation or response to gradients in auxin across a root tip needed for gravity response (Rashotte et al. 2001). We are now currently investigating other tissues to determine if there are altered gravity responses in the hypocotyl and inflorescences of Arabidopsis. The hypocotyls of rcn1, when grown in the dark, are phenotypically different from wild-type. Like roots, rcn1 hypocotyls have elevated rates of auxin transport (Muday et al. in review). But in opposition to roots, this elevated auxin transport is correlated with an enhanced gravitropic response. We have also found that the rcn1 phenotype is only present in seedlings grown in the dark (Muday et al. in review), suggesting an interesting intersection between photomorphogenesis and auxin transport.<br><br>Ethylene regulation of gravitropic curvature:<br><br>Our experiments both in phosphorylation and flavonoid control of auxin transport during gravitropism have led us to identify connections between auxin transport and the ethylene signal transduction pathway. First, the rcn1 mutant was isolated in a screen for altered response to ethylene and found to have elevated ethylene synthesis. We have examined the relationship between elevated auxin transport, reduced hypocotyl elongation, elevated gravitropism, and ethylene response in rcn1 and in double mutants between rcn1 and plants with mutations in genes required for ethylene signaling. Our results suggest that the rcn1 mutation does not alter the response of hypocotyls to ethylene during short term exposure, but that part of the dark dependent phenotype is an elevated ethylene synthesis that inhibits hypocotyl elongation (Muday et al. in review). These studies find that the ein2 mutant, which is ethylene insensitive, has a profound delay in gravitropic response. Additionally, elevated levels of ethylene inhibit hypocotyl gravitropism, consistent with a requirement of ethylene for gravitropism, but an inhibitory role at high levels of ethylene. Yet, the rcn1 mutation, which enhances hypocotyl gravitropism, when combined with the ein2 mutant, results in plants with an enhanced gravity phenotype. Our conclusions are that the RCN1 protein likely acts in parallel in regulation of |

dephosphorylation of two separate pathways. These results are consistent with a complex interplay in signaling between auxin transport and ethylene in control of hypocotyl gravitropism (Muday et al. in review).

Our studies of the flavonoids and gravity response have also converged with ethylene signaling. In Arabidopsis roots, ethylene also inhibits root gravitropism. The roots of tt4 are greatly reduced in their sensitivity to exogenous ethylene (Buer et al. in review), which suggested the idea that ethylene might affect root gravitropism by altering flavonoid accumulation. Consistent with this prediction, flavonoid accumulation is elevated in plants with elevated ethylene and this accumulation is absent in the ethylene insensitive, ein2 mutant. Together these results are consistent with ethylene inhibition of root gravitropism being mediated by enhanced flavonoid accumulation, which then reduces root auxin transport (Buer et al., in review).

IBA transport and gravitropic response:

A collaborative project that has been supported by this grant is a comparison of transport of two different endogenous auxins, IAA and IBA, in Arabidopsis. We have found that there are significant differences in the transport of these two auxins, with IBA transport being unchanged in mutants that alter IAA transport and in response to IAA efflux inhibitors, consistent with transport of this auxin being mediated by a separate set of proteins (Rashotte et al. 2003). We have also characterized auxin transport in the rib1 mutant, which was isolated based on its altered response to exogenous IBA. We have found that this mutant has reduced IBA transport, but wild-type levels of IAA transport (Poupart et al., 2005). These findings link transport of the two auxins IBA and IAA to gravitropic response in Arabidopsis roots.

Other collaborative projects:

Several other collaborative projects have been supported by this grant. In collaboration with the laboratory of Dr. Stan Roux, who has a project examining the effect of exogenous ATP on gravity response, we have asked whether ATP is blocking gravity response through inhibition of auxin transport. Our results support this possibility, as auxin transport and gravity response are reduced in parallel by these treatments (Tang et al. 2002). We have continued our collaboration with Dr. Sarah Wyatt, to examine auxin transport in her gps mutants, which have altered gravitropic signaling (Nadella et al, in press). We find no evidence of altered polar auxin transport in these mutants, but lateral auxin transport appears to correlate with the gravity defects in these mutants. In collaboration with Dr. Zhenbaio Yang, we have examined auxin transport in response to altered levels of the plant hormone, brassinosteroid. This work has suggested a link between IAA and brassinosteroids (Bao et al. 2004). We have also collaborated with Dr. Leslie Sieberth on characterization of the scarface mutant, which has a defect in vein patterning in leaves to show that there is altered auxin transport in the roots of this mutant (Sieberth et al., in review). Two collaborative review articles were also written with support of this grant (Muday and Murphy, 2002; Muday et al. 2003).

| Bibliography Type: | Description: (Last Updated: 08/25/2020) |
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